

Assignment 6

Simulating and Analyzing a Discrete Random Variable

Imagine a fair six-sided die. The outcomes when rolling the die are values from 1 to 6 — each outcome is a value of a discrete random variable.

Sub-Questions:

(a) Simulate a Discrete Random Variable

1. Simulate 500 rolls of a fair die using `random.randint(1, 6)` or `numpy.random.randint(1, 7, size=500)`.
2. Display the first 10 outcomes.

(b) Frequency and Visualization

1. Count how many times each number (1 to 6) appears.
2. Plot a bar chart showing the frequency of each outcome.
3. Which number appeared the most? The least?

(c) Estimating Probabilities

1. Compute the relative frequency (frequency/total) for each outcome.
2. Are these probabilities close to the expected value of $1/6$?
3. Repeat the experiment with 5000 rolls. Are the estimated probabilities now closer to $1/6$?